

PCR-5D3

Successor-Target Quantitative Active Closure of the Neutrino Standing-Carrier Space

Finite-Operational Process Normalization, RCF-Finite Codescent,
and Proper-Continuum Closure

Target scope: THUCL-D3-1.3

Target-specific scope

The results proved here apply to the successor target THUCL-D3-1.3. The separately frozen THUCL-D3-1.1 Global Fibre Completion problem is neither completed nor revised. Although the two programmes share upstream PCR-3, PCR-4, PCR-5B, PCR-5C, and PCR-5D3 objects, conclusions that use the 1.3 quantitative coverage rule, retained process tester, or finite-operational comparison class do not transport backward across the target boundary.

Abstract

This paper determines the action of the successor heavy-state, active-closure, weak-universality, and charged-lepton-flavour target THUCL-D3-1.3 on the exact neutrino-oscillation standing-carrier space. The successor target is kept mathematically separate from the original THUCL-D3-1.1 Global Fibre Completion programme. Its defining quantitative coordinate is the retained electron-active process-tester contraction

$$C_{ee}^{\text{op},1.3}(X) = W_X * \mathcal{T}_{e,\text{act}}^{U,\text{ret}}.$$

Every finite-operational law admits a unique positive process-comb representation. Deterministic restriction of the D3-only slots then proves the retained-base identity

$$C_{ee}^{\text{op},1.3}(Y) = \overline{C}_{ee}^{\text{osc}}(\rho_{D3}^{1.3}[Y]),$$

so the quantitative normalization is constant on each complete lift fibre. On the finite-unitary submatrix continuum $q_{\text{sub}}(\delta)$, $\delta \in [0.02, 0.03]$, its value is $1 - \delta \leq 0.98 < 0.987$, and every corresponding 1.3 fibre is empty. A compact prompt-HNL continuum remains accepted. Consequently, in the full finite-operational target scope,

$$\emptyset \subsetneq \text{im } \rho_{D3}^{1.3} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \quad |\text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}, \quad |Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}.$$

The successor target therefore has a proper continuum restriction image and does not license a unique carrier realization.

The global source-atlas theorem is proved on the independently defined RCF-finite semantic subcategory. Its source normal form is invariant under lawful source and parameter transport while retaining every active bearer, derivation hyperedge, support incidence, constitutive relation, and ordered algebraic operation. Source provenance is retained without becoming a second terminal standing classifier. Presentation overlap is represented by a genuine source-indexed Čech nerve over the standing core, whereas physical source amalgamation requires an independently supplied typed gluing diagram. The complete joint residual family is generated naturally from source-domain conditions, constitutive equations, process composition and interference, common-apex compatibility, and typed updates.

Each finite residual problem reduces to a first-order formula over the real algebraic numbers. Replacing every algebraic coefficient by its rational minimal-polynomial and isolating-interval data gives an exact formula over $\mathbb{Q}$. Standard quantifier elimination for real closed fields and semialgebraic projection then supply satisfiability, projection, cell transport, and stabilizer conjugacy. Noncircular chartification followed by groupoid codescent reconstructs the RCF-finite semantic category and yields

$$\text{Status}_{\text{supp}}(D3, 1.3; \text{RCF} - \text{fin}) = \text{ClosureCertified}_{\text{RCF} - \text{fin}}^{1.3}.$$

The remaining finite-operational but non-RCF-finite sector retains status $\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF} - \text{fin}}^{1.3}$, not elimination. Computational materials accompany the proof as reproducibility, provenance, typing, and formula-formation checks; they are not substitutes for the semantic reduction or the standard real-closed-field theorem.

Keywords. AASC; neutrino oscillation; target extension; process tensor; quantum comb; standing carrier; heavy neutral lepton; nonunitarity; charged-lepton flavour violation; real closed fields; quantifier elimination; Čech codescent.

Principal theorem

Let $T_{D3}^{\text{HUL},1.3}$ be the target-complete successor object formed from TH-D3-1, TU-D3-1.3, and TL-D3-1, with restriction

$$R_{D3}^{1.3} : \mathcal{N}_{D3}^{1.3} \longrightarrow \mathcal{N}(T_{\text{osc}}), \quad \rho_{D3}^{1.3} = \pi_0(R_{D3}^{1.3}), \quad \mathcal{L}_{D3}^{1.3}(q) = \mathcal{N}_{D3}^{1.3} \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

Two conclusions are established.

Finite-operational endpoint. The retained electron-active tester factors through the oscillation standing class. The prompt-HNL continuum lies in the restriction image,

while the finite-unitary submatrix continuum lies in its complement. Hence

$$\emptyset \subsetneq \text{im } \rho_{D3}^{1.3} \subsetneq Q_{\text{osc}}^{\text{SCIC}}, \quad |\text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}, \quad |Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}.$$

The successor target therefore remains non-singleton at terminal carrier resolution. *RCF-finite global closure.* On $\mathbf{QProc}_{D3}^{\text{RCF-fin}} = \mathbf{QProc}_{D3}^{\text{RCF-fin}}$, the B15-framed source normal form, standing-core/provenance separation, actual-index Čech nerve, typed physical-amalgamation category, transport-natural residual family, exact real-algebraic coefficient descent, semantic chartification, and groupoid codescent give

$$\mathcal{L}_{D3}^{1.3, \text{RCF-fin}}(q) \simeq \text{codesc} \mathcal{U}_{D3, \bullet}^{1.3, \text{RCF}}(q), \quad \text{codesc} \mathcal{U}_{D3, \bullet}^{1.3, \text{RCF}} \simeq \mathcal{N}_{D3, \text{sem}}^{1.3, \text{RCF-fin}},$$

and therefore

$$\text{GFC-09B}^{1.3, \text{exact-fin}}, \quad \text{GFC-25}^{1.3, \text{exact-fin}}, \\ \text{Status}_{\text{supp}}(D3, 1.3; \text{RCF-fin}) = \text{ClosureCertified}_{\text{RCF-fin}}^{1.3}.$$

The parent remainder $\mathbf{QProc}_{D3}^{\text{fin-op}} \setminus \mathbf{QProc}_{D3}^{\text{RCF-fin}}$ has global-atlas status $\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$. None of these conclusions is asserted for THUCL-D3-1.1.

Why the target distinction is load-bearing

Under TU-D3-1.3, the retained electron-active tester is a universal quantitative acceptance coordinate. This converts the finite-unitary deficit into a complete-fibre obstruction. Under TU-D3-1.1, the same calculation remains a marginal U-chart obstruction. The difference is therefore part of the scientific target, not a change of presentation.

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1 Introduction and separation of target programmes

The PCR-5D3 problem asks how heavy neutral states, active-sector closure, weak-universality information, and charged-lepton-flavour observables act on the already classified neutrino-oscillation standing-carrier space. The decisive issue is target identity. The original target THUCL-D3-1.1 and the successor target THUCL-D3-1.3 share the same upstream carrier and restriction structures, but they impose different universal acceptance semantics. They therefore define different fibre problems.

The present analysis has two logically independent conclusions. First, at the full finite-operational level, retained-base factorization makes the 1.3 quantitative normalization constant on every lift fibre. This produces a continuum of empty fibres and, independently, a continuum of accepted prompt-HNL fibres. The restriction image is thus a proper continuum subset of $Q_{\text{osc}}^{\text{SCIC}}$. Second, on the RCF-finite semantic subcategory, source normalization, typed overlap, physical amalgamation, residual naturality, exact coefficient descent, chartification, and groupoid codescent close the global atlas. The first conclusion does not depend on the second.

The proof is organized accordingly. Sections 3–6 establish the finite-operational representation, retained tester, and complete-fibre obstruction. Sections 7–14 construct the source-indexed RCF-finite atlas and its codescent. Sections 15–18 give the surviving continuum, the proper-continuum endpoint, the closure theorem, and the role-occupancy consequence. The final sections state the mathematical/computational authority boundary and the exact conditions under which the result would reopen.

1.1 The original Global Fibre Completion target

The accepted PCR-5D3 publication and the approved completion skeleton fix

$$T_{D3}^{\text{HUL},1.1} = \text{THUCL-D3-1.1}.$$

That programme asks for source, family, fibre, overlap, support, and residue completion with the original U-pass semantics held unchanged. Its own skeleton states that a target update is a new certificate rather than completion of the old target. The 1.1 problem therefore remains exactly where the approved skeleton places it.

1.2 The successor active-closure target

The successor line has the distinct target history

$$\text{THUCL-D3-1.1} \longrightarrow \text{THUCL-D3-1.2} \longrightarrow \text{THUCL-D3-1.3}.$$

The first arrow added quantitative coverage and a universal operational-normalization threshold. The second replaces the frozen one-step-channel QNorm definition with a process-tensor tester contraction and explicitly narrows the process comparison class. These arrows preserve prior truths but do not transport later conclusions backward.

Theorem 1.1 (STX-00: Programme-separation theorem). *The 1.1 Global Fibre Completion programme and the 1.3 successor-target programme share the upstream PCR-3, PCR-4, PCR-5B, PCR-5C, and published PCR-5D3 base objects, but diverge at the target certificate. A theorem whose proof uses QCov, the universal QNorm threshold, the retained tester of TU-D3-1.3, or the finite-operational process-scope rule is a theorem of Programme B only.*

Proof. Target identity includes the admitted records, applicability rules, tester family, acceptance semantics, comparison class, and update boundary. Each listed item differs

between 1.1 and 1.3. PCR-5C target-extension discipline and the fixed-domain no-repair rule therefore classify the change as a new certificate. No implication from 1.3 acceptance to 1.1 completion, or conversely, follows merely from shared upstream objects. $\square$

Corollary 1.2 (Theorem-label discipline). *Unqualified labels GFC-09B, GFC-25, and ClosureCertified remain reserved for the 1.1 programme. This manuscript uses only scope-qualified analogues $\text{GFC-09B}^{1.3,\text{exact-fm}}$, $\text{GFC-25}^{1.3,\text{exact-fm}}$, and $\text{ClosureCertified}_{\text{RCF-fm}}^{1.3}$.*

Status 1.3 (Mini-ARC status). No Mini-ARC update is made. A future decision to substitute 1.3 for the currently planned D3 target would require a separate ARC target-update memorandum. The accompanying programme memorandum explicitly records this non-supersession.

2 Target, base space, and inherited structure

2.1 Oscillation base and B15 source frames

PCR-5B supplies the exact oscillation groupoid atlas

$$\mathcal{A}_{\text{osc}}^{\text{exec}} \simeq \mathcal{N}_{5,\text{osc}}^{\text{exec}}, \quad \pi_0 \mathcal{A}_{\text{osc}}^{\text{exec}} \cong Q_{\text{osc}}^{\text{SCIC}}, \quad |Q_{\text{osc}}^{\text{SCIC}}| \geq \mathfrak{c},$$

with complete standing packet

$$\Xi_{\text{osc}} = (T^{\text{face}}, R^{\text{stand}}, \Delta_{\text{fr}}^{\text{del}}, C^{\text{act}}, M^{\text{match}}, B^{\text{bdy}}, F^{\text{same}}).$$

For every PCR-3 family and generated source skeleton $a = (F, \sigma)$, the frozen B15 datum contains its exact raw source domain, equations, applicability, total status, PCR-4 solution object, lawful presentation groupoid, concrete core functor, solved packet moduli, framed provenance, failure frontier, separator set, and overlap data. These objects remain the exact base data for the successor calculation; no finite model list replaces them.

2.2 The successor target

The accompanying addendum defines

$$T_U^{1.3} = (T_U^{1.2} \setminus T_U^{\text{QNorm},1.2}) \cup T_U^{\text{QNorm},1.3} \cup T_U^{\text{ProcScope},1.3},$$

and

$$T_{D3}^{\text{HUL},1.3} = T_H^{1.0} \cup T_U^{1.3} \cup T_L^{1.0}.$$

It inherits the 1.2 quantitative registry, guards, covariances, likelihoods, status semantics, complete-coverage rule, and threshold $c_e^{\min} = 0.987$. Only the QNorm definition and the process comparison class differ from the 1.2 parent target.

Theorem 2.1 (STX-01: Target/source/continuation lock). *The PCR-3 source ontology, PCR-4 one-apex physicality, PCR-5B/B15 base atlas, PCR-5C restriction calculus, H and L target records, U quantitative registry, QCov rule, QNorm tester, process comparison class, threshold, and continuation monoid are fixed before the fibre calculation. No source family, deficit interval, prompt interval, survivor, empty fibre, image cardinality, or desired endpoint enters target formation.*

Proof. The target certificate is formed only from prior target records, the old retained active/null tester, deterministic extension along the target embedding, and the finite-operational axioms. The source registry and endpoint calculation appear downstream. The separate target addendum contains no source interval or fibre result, which establishes the required dependency order. $\square$

3 Finite-operational process representation

3.1 Operational laws

Fix finitely many causally ordered intervention slots. At slot j , let H_j^{in} and H_j^{out} be finite-dimensional Hilbert spaces and let Ω_j be a standard-Borel outcome space. A normal instrument is a countably additive map from Borel sets to completely positive trace-nonincreasing maps whose total is trace preserving.

Definition 3.1 (Finite-operational law). A finite-operational law is a joint probability assignment

$$\omega(\mathcal{A}_{n-1}(B_{n-1}), \dots, \mathcal{A}_0(B_0))$$

that is multilinear under finite instrument mixing, completely positive under arbitrary finite ancillary extension, countably additive in every Borel event, bounded, normalized on deterministic instruments, and causal: replacing future instruments by deterministic totals leaves earlier marginals unchanged.

Definition 3.2 (Finite-operational process category). $\mathbf{QProc}_{D3}^{\text{fin-op}}$ has finite-operational laws as objects. Arrows are typed normal supermaps and process/tester isomorphisms preserving slot order, causal cuts, boundary reference, Borel outcome structure, and deterministic tester totals. Continuous, functional, and categorical moduli may occur in the parameter object; each instantiated quantum operation space remains finite-dimensional.

3.2 Representation theorem

Theorem 3.3 (STX-02: Finite-operational process-tensor representation). *Every finite-operational law ω is represented by a unique positive process-comb operator W_ω on the tensor product of the finite-dimensional Choi operation spaces such that*

$$\omega(A_{n-1}, \dots, A_0) = W_\omega * (A_{n-1} \otimes \dots \otimes A_0).$$

The causal law is equivalent to the recursive comb normalization conditions

$$\text{Tr}_{\text{out}_k} W^{(k)} = I_{\text{in}_k} \otimes W^{(k-1)}$$

at every causal cut, with the initial scalar normalization at the first slot. Conversely, every positive operator satisfying these conditions defines a unique finite-operational law. For standard-Borel instruments, the resulting event map is countably additive.

Proof. For each slot, the Choi operators of completely positive maps span a finite-dimensional complex vector space V_j . Multilinearity uniquely linearizes ω to a functional on $V_{n-1} \otimes \dots \otimes V_0$. The Hilbert-Schmidt duality of this finite-dimensional tensor product gives a unique representing operator W_ω . Complete positivity under ancillary extension is exactly positivity of the representing operator on matrix-positive operation tensors, hence $W_\omega \geq 0$. Normalization on deterministic instruments and causal invariance under future deterministic replacement yield the recursive partial-trace constraints by testing against a spanning set of earlier operations. Conversely, a positive operator with those constraints produces nonnegative normalized probabilities for every instrument sequence and has the required causal marginals. For a standard-Borel instrument, each event operator is countably additive; the finite-dimensional continuous linear functional induced by W_ω preserves that countable additivity. Uniqueness follows because the instrument Choi operators span every V_j . $\square$

Remark 3.4 (Why the category is narrowed). No infinite-dimensional Choi or predual representation theorem is assumed. A source with unbounded intervention algebras, genuinely infinite quantum slots, nonnormal outcome dependence, or no finite target-facing operational compression is not silently included. The successor endpoint is exact relative to $\mathbf{QProc}_{D3}^{\text{fin-op}}$, which is part of the target certificate.

Definition 3.5 (Process-scope status). Every source-coherent proposal receives one of

$\text{Pass}_{\text{Proc}}, \text{ROOS}_{\text{Proc}}, \text{Undef}_{\text{Proc}}, \text{SourceUpdate}, \text{TargetUpdate}.$

The first requires a constructed object of $\mathbf{QProc}_{D3}^{\text{fin-op}}$. $\text{ROOS}_{\text{Proc}}$ means physically meaningful but outside the finite-operational comparison class. $\text{Undef}_{\text{Proc}}$ means an asserted in-scope law fails positivity, boundedness, measurability, or causal normalization. New source sorts and new target slots/testers route to updates.

Theorem 3.6 (STX-03: Total process-scope disposition). *Every finite activity-reduced PCR-3/B15 source proposal has exactly one process-scope disposition before target acceptance. No $\text{ROOS}_{\text{Proc}}, \text{Undef}_{\text{Proc}}, \text{SourceUpdate}$, or TargetUpdate point is counted as an accepted 1.3 lift.*

Proof. The finite source grammar determines whether the claimed target-facing operations and outcome spaces are already typed. If they are typed and satisfy the finite-operational axioms, STX-02 constructs the unique comb. If they are physically meaningful but require a different operational category, the point is response-out-of-scope. A claimed in-scope violation is undefined. Missing source or target primitives are updates. These cases are disjoint by their formation conditions and exhaustive relative to the declared comparison class. $\square$

4 Source laws and physical process realizations

4.1 Faithful source category

Definition 4.1 (Source-law category and lawful parameter-transport groupoid). $\mathbf{Src}_{D3}^{\text{coh}}$ has as objects the finite activity-reduced PCR-3/B15 source hypergraphs together with the source-semantic part of their B15 frames: source-parameter declarations and domains, constitutive equations and inequalities, source incidences and external bearers, and the declared source/parameter transport data. Source data may include primitive or derived nodes, finite incidences and attachments, Hamiltonian or interaction generators, bounded GKSL generators, transition amplitudes, memory kernels, correspondences, projections, binding or fusion maps, finite higher relations, and already completely positive instruments.

A morphism is a pair $\psi = (\phi, \psi_\Theta)$, where ϕ is a lawful typed source-node isomorphism and $\psi_\Theta : \Theta \rightarrow \Theta'$ is an exact invertible semialgebraic parameter transport over $\mathbb{R}_{\text{alg}}$. The forward and inverse coordinate maps are finite typed semialgebraic expressions; their inverse laws are proved exactly, directly for affine transports and by an RCF certificate for general semialgebraic transports. They must preserve the complete activity-reduced carrier trace and carry the source-equation locus to the source-equation locus. Composition is typed substitution and the verified inverse is the groupoid inverse.

Parameter alpha-renamings form the identity-bearing subgroupoid. They carry no independent source standing and are collapsed by canonical labeling. A nontrivial lawful transport—for example an affine chart change $y = x - 1$ —is not collapsed to equality of presentations. It is retained as a canonical normalized arrow between the corresponding framed source normal forms.

A source generator is not retyped as a completely positive map. Complete positivity belongs to the realized operational process. A Hamiltonian law is retained as a generator represented by its finite-time unitary process; an amplitude is retained as a source transition block represented by a normalized instrument only on its physical contraction locus; a memory law is retained through the multi-time process equations it imposes.

Definition 4.2 (Realization groupoid). For $H \in \mathbf{Src}_{D3}^{\text{coh}}$, let $\mathfrak{R}_{D3}(H)$ be the groupoid of triples

$$(W_H, \eta_H, \mathfrak{a}_H),$$

where $W_H \in \mathbf{QProc}_{D3}^{\text{fin-op}}$, η_H represents every active source law and incidence in W_H , and $\mathfrak{a}_H$ is one PCR-4 physical common-apex witness satisfying the inherited and D3 residual equations. Arrows are source-framed process isomorphisms commuting with η_H , the target testers, and the base frame. Define

$$\mathbf{Real}_{D3}^{\text{fin-op}} := \int_{H \in \mathbf{Src}_{D3}^{\text{coh}}} \mathfrak{R}_{D3}(H).$$

Theorem 4.3 (STX-04: Source-law realization and exact reduct). *The projection*

$$U_{D3} : \mathbf{Real}_{D3}^{\text{fin-op}} \longrightarrow \mathbf{Src}_{D3}^{\text{coh}}, \quad (H, W_H, \eta_H, \mathfrak{a}_H) \longmapsto H,$$

is exact. Every in-scope semantic D3 physical realization is represented by an object of $\mathbf{Real}_{D3}^{\text{fin-op}}$, and every object of $\mathbf{Real}_{D3}^{\text{fin-op}}$ is an in-scope semantic D3 realization. Empty realization fibres and non-Pass process statuses are explicit dispositions rather than omissions from the source grammar.

Proof. PCR-3 gives every finite same-target carrier architecture an activity-reduced source normal form. A semantic in-scope operational realization supplies a finite-operational probability law on the target slots; STX-02 gives its unique comb W_H . The source equations and active incidences define η_H , and PCR-4 one-point physicality supplies $\mathfrak{a}_H$. Conversely, an object of $\mathbf{Real}_{D3}^{\text{fin-op}}$ is a normalized causal operational process satisfying the complete source equations and one-apex physical constraints, which is precisely an in-scope semantic realization. The source object remains the first component of the Grothendieck construction, so no generator or incidence is replaced by a downstream process label. $\square$

5 Operational normalization by a retained process tester

5.1 Retained tester

The old oscillation target contains an event-complete source-propagation-detection instrument with active charged-current outcomes and a null/hidden outcome. Coarsen the active electron outcomes into the positive tester $\mathcal{T}_{e,\text{act}}^{\text{osc}}$, and let $\mathcal{T}_{e,\perp}^{\text{osc}}$ be its complement. Their sum is the deterministic electron tester.

Let $j : T_{\text{osc}} \hookrightarrow T_{D3}^{\text{HUL},1.3}$ be the target embedding. The 1.3 target freezes the deterministic extension

$$\mathcal{T}_{e,\text{act}}^{U,\text{ret}} = j! \mathcal{T}_{e,\text{act}}^{\text{osc}},$$

obtained by filling every D3-only slot with its deterministic identity/discard operation. Define the complementary extension similarly.

Definition 5.1 (Universal operational normalization). For an in-scope D3 physical point X with process comb W_X , define

$$C_{ee}^{\text{op},1.3}(X) = W_X * \mathcal{T}_{e,\text{act}}^{U,\text{ret}}.$$

This contraction is the definition; a one-step channel is not assumed.

Theorem 5.2 (STX-05: Positivity, normalization, covariance, and additivity). *For every in-scope X ,*

$$0 \leq C_{ee}^{\text{op},1.3}(X) \leq 1, \quad C_{ee}^{\text{op},1.3}(X) + C_{e\perp}^{\text{op},1.3}(X) = 1.$$

Lawful simultaneous process/tester transport leaves both scalars invariant. The event map obtained from a standard-Borel active-event decomposition is countably additive.

Proof. The process comb and both tester effects are positive. Their link products are therefore nonnegative. Linearity and deterministic tester normalization give

$$W_X * (\mathcal{T}_{e,\text{act}}^{U,\text{ret}} + \mathcal{T}_{e,\perp}^{U,\text{ret}}) = W_X * \mathcal{T}_e^{U,\text{det}} = 1,$$

which bounds each summand by one. A lawful arrow acts contragrediently on the process and tester legs, so the internal isomorphisms cancel in the scalar contraction. Countable additivity follows from that of the instrument-valued measure and finite-dimensional continuity of the process functional. $\square$

Theorem 5.3 (STX-06: Factorizing-branch reduction). *If the relevant slots factorize as an electron preparation ρ_e , a normal CPTP map Φ_X , and an active detector effect $E_{e,\text{act}}$, with deterministic operations on all other slots, then*

$$C_{ee}^{\text{op},1.3}(X) = \text{Tr}[E_{e,\text{act}}\Phi_X(\rho_e)].$$

Proof. The process Choi operator and tester factor into preparation, channel, and effect legs. Associativity of the link product first contracts the preparation into the channel and then the channel output into the effect, yielding the ordinary Born-rule trace. No such reduction is used on a correlated or memory-bearing branch. $\square$

6 The retained-base factorization theorem

6.1 Old active-event standing

Definition 6.1 (Old active-event functional). For a concrete old oscillation process $x \in \mathcal{N}(T_{\text{osc}})$, set

$$\overline{C}_{ee}^{\text{osc}}(x) = W_x * \mathcal{T}_{e,\text{act}}^{\text{osc}}.$$

Theorem 6.2 (STX-07: Old active-event witness factors through SCIC). *The map $\overline{C}_{ee}^{\text{osc}}$ is a target-admitted standing use and descends to a well-defined function*

$$\overline{C}_{ee}^{\text{osc}} : Q_{\text{osc}}^{\text{SCIC}} \longrightarrow [0, 1].$$

Proof. The PCR-2 target-facing trace and the PCR-5B target fingerprint retain the charged-current boundary, the complete event instrument, and its active/null outcome semantics. The coarsened electron-active probability is therefore a functional of the frozen target-face component of Ξ_{osc} . PCR-5A/SCIC packet completeness states that equal SCIC classes have equal values under every admitted standing continuation witness. Hence the scalar is constant on each SCIC class and descends to the quotient. $\square$

6.2 Deterministic restriction identity

For an enriched process Y , PCR-5C candidate restriction and the 1.3 process-scope rule define the old process by deterministic contraction of all D3-only slots:

$$W_{R_{D3}^{1.3}(Y)} = W_Y * \mathcal{T}_{D3 \setminus \text{osc}}^{\text{det}}.$$

The retained tester is the old tester tensored/linked with the same deterministic operations.

Theorem 6.3 (STX-08: Retained-base QNorm factorization). *For every accepted in-scope enriched physical point Y ,*

$$C_{ee}^{\text{op},1.3}(Y) = \overline{C}_{ee}^{\text{osc}}(R_{D3}^{1.3}Y).$$

After quotient descent,

$$C_{ee}^{\text{op},1.3}([Y]) = \overline{C}_{ee}^{\text{osc}}(\rho_{D3}^{1.3}[Y]).$$

Consequently, for every base class q , the QNorm function is constant on the entire fibre $\mathcal{L}_{D3}^{1.3}(q)$:

$$C_{ee}^{\text{op},1.3}|_{\mathcal{L}_{D3}^{1.3}(q)} \equiv \overline{C}_{ee}^{\text{osc}}(q).$$

Proof. By definition of deterministic target extension,

$$\mathcal{T}_{e,\text{act}}^{U,\text{ret}} = \mathcal{T}_{e,\text{act}}^{\text{osc}} * \mathcal{T}_{D3 \setminus \text{osc}}^{\text{det}}.$$

Associativity of the link product and comb causality give

$$\begin{aligned} W_Y * \mathcal{T}_{e,\text{act}}^{U,\text{ret}} &= W_Y * (\mathcal{T}_{e,\text{act}}^{\text{osc}} * \mathcal{T}_{D3 \setminus \text{osc}}^{\text{det}}) \\ &= (W_Y * \mathcal{T}_{D3 \setminus \text{osc}}^{\text{det}}) * \mathcal{T}_{e,\text{act}}^{\text{osc}} \\ &= W_{R_{D3}^{1.3}(Y)} * \mathcal{T}_{e,\text{act}}^{\text{osc}}. \end{aligned}$$

STX-07 makes the right side representative independent. If two enriched points lie in the same homotopy fibre, their restrictions lie in the same connected old SCIC core, so both have the same value. Any enriched source term that changes the value changes the restricted old process and therefore exits that fibre; it is not compensation inside the fibre. $\square$

Structural consequence

This theorem discharges the exact properness boundary. QNorm is not merely declared “retained.” Its process-tensor definition is proved to be the deterministic extension of an already standing old tester, and its value is therefore mathematically forced on every same-base lift.

7 A finite-unitary continuum and complete-fibre exclusion

7.1 Base calibration

Let U_ν be the frozen old visible unitary and $\delta \in [0.02, 0.03]$. The accepted PCR-5B base calibration uses the finite unitary dilation

$$U_\delta = \begin{pmatrix} \sqrt{1-\delta} U_\nu & \sqrt{\delta} I_3 \\ -\sqrt{\delta} U_\nu & \sqrt{1-\delta} I_3 \end{pmatrix}, \quad N_\delta = \sqrt{1-\delta} U_\nu.$$

Its event-complete instrument is

$$\mathcal{E}_{\lambda,\beta}^{(\delta)} = (1 - \delta)\mathcal{E}_{\lambda,\beta}^0,$$

for active outcomes, with the missing probability transferred to the null/hidden outcome. The conditionally normalized visible kernel remains the old PMNS kernel.

Theorem 7.1 (STX-09: Exact base value on the submatrix continuum). *For the retained base class $q_{\text{sub}}(\delta)$,*

$$\overline{C}_{ee}^{\text{osc}}(q_{\text{sub}}(\delta)) = 1 - \delta.$$

The map $\delta \mapsto q_{\text{sub}}(\delta)$ is injective in $Q_{\text{osc}}^{\text{SCIC}}$; hence $|\mathcal{Q}_{\text{sub}}| = \mathfrak{c}$, where

$$\mathcal{Q}_{\text{sub}} = \{q_{\text{sub}}(\delta) : \delta \in [0.02, 0.03]\}.$$

Proof. On this lossless active calibration branch the old active tester sums the three active charged-current outcomes. Therefore

$$\overline{C}_{ee}^{\text{osc}}(q_{\text{sub}}(\delta)) = \sum_{\beta} \text{Tr}[\mathcal{E}_{\lambda,\beta}^{(\delta)}(\rho_e)] = (1 - \delta) \sum_{\beta} \text{Tr}[\mathcal{E}_{\lambda,\beta}^0(\rho_e)] = 1 - \delta.$$

Different δ values give different values of a retained standing witness, so SCIC packet completeness forbids their identification. A nondegenerate real interval has cardinality $\mathfrak{c}$. $\square$

Theorem 7.2 (STX-10: Complete-fibre QNorm obstruction). *For every $\delta \in [0.02, 0.03]$,*

$$C_{ee}^{\text{op},1.3} \big|_{\mathcal{L}_{D3}^{1.3}(q_{\text{sub}}(\delta))} \equiv 1 - \delta \leq 0.98 < 0.987.$$

Hence

$$\mathcal{L}_{D3}^{1.3}(q_{\text{sub}}(\delta)) = \emptyset.$$

Proof. STX-08 fixes the QNorm value on every same-base lift, and STX-09 evaluates the base witness. The QNorm chart is applicable to every in-scope point with the frozen electron boundary and is a mandatory 1.3 acceptance relation. Every point in the putative fibre therefore fails the same chart. An added source term cannot raise the scalar while remaining in the fibre because any change in the scalar changes the old retained process. The inherited QCov rule may independently fail uncovered U-active primitives, but it is not needed for this strict numerical obstruction. $\square$

Corollary 7.3 (Continuum complement).

$$\mathcal{Q}_{\text{sub}} \subseteq Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{D3}^{1.3}, \quad |Q_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}.$$

Nonclaim 7.4. STX-10 is not a theorem under THUCL-D3-1.1. The older target did not contain the universal 1.3 QNorm acceptance relation, so its finite-submatrix result remains marginal.

8 The actual-index source atlas

8.1 Dispatch types are not chart indices

Every base index $a = (F, \sigma)$ continues to range over the complete PCR-5B/B15 source grammar. The twelve local D3 labels remain useful as a finite dispatch codomain:

ID	Dispatch meaning	What the label does not identify
H1-H5	direct, hidden, chain, higher, and source-null H response	a B15 source frame, source term, parameter cell, process, or apex
U1-U3	open/memory, finite-block, and boundary U response	a unique microscopic source or standing carrier
L1-L3	direct, loop/interference, and source-null L response	a complete source provenance class
J1	finite mixed or irreducibly higher H/U/L response	a single universal source object or a finite model list

Write $\mathcal{D}_{12}$ for this discrete dispatch set. It is the codomain of a routing functor, not the atlas index.

Definition 8.1 (Actual chart index). An actual target-qualified chart index is a tuple

$$i = (a, \tau, H, \theta, W_H, \alpha_H, \kappa_H),$$

where:

1. $a = H_{F,\sigma}$ is one concrete frozen B15 source frame, including its source skeleton, raw parameter object, source equations, applicability/status object, lawful presentation groupoid, chart-to-core functor, framed provenance, separator set, and failure frontier;
2. $\tau \in \mathcal{D}_{12}$ is only the response-dispatch type;
3. H is the actual finite activity-reduced PCR-3/D3 source constructor term over a ;
4. θ is the exact source/process parameter cell on which the chart equations and status guards are fixed;
5. W_H is the solved finite-operational process record and its comb representation certificate;
6. α_H is the one-point PCR-4/D3 physical-apex record, including all cross-incidence and common-cone data; and
7. κ_H is the complete proof-carrying chart certificate: source coherence, process representation, retained-base restriction, applicable target status/value equations, matching, boundary, continuation, physicality, and lawful action.

Let $\mathcal{J}_{D3}^{1.3}$ denote the category of these indices and lawful framed chart arrows. The routing functor

$$\text{disp} : \mathcal{J}_{D3}^{1.3} \longrightarrow \mathcal{D}_{12}$$

forgets all actual source/process data except τ .

Remark 8.2 (Dispatch cardinality is not atlas cardinality). The numbers $12^2 = 144$ and $\sum_{r=3}^{12} \binom{12}{r} = 4017$ count combinations of dispatch labels only. They do not count objects, pairs, or higher simplices of $\mathcal{J}_{D3}^{1.3}$, because distinct B15 frames and distinct source presentations may share the same dispatch value. No atlas theorem below depends on these formal label counts.

8.2 Independent RCF-finite semantic scope

The closed global-atlas scope is not defined by possession of a chart certificate. Fix the effective real closed coefficient field

$$\mathbb{R}_{\text{alg}} = \{x \in \mathbb{R} : x \text{ is algebraic over } \mathbb{Q}\},$$

represented by square-free integer polynomials, canonical isolating rational intervals, and root indices. Ordered-field arithmetic, equality, and sign determination are effective.

Definition 8.3 (RCF-finite semantic process scope). Let

$$\mathbf{QProc}_{D3}^{\text{RCF-fin}} := \mathbf{QProc}_{D3}^{\text{RCF-fin}} \subset \mathbf{QProc}_{D3}^{\text{fin-op}}$$

be the full semantic subcategory of accepted THUCL-D3-1.3 finite-operational physical realizations satisfying all of the following physical and operational conditions:

1. the participating carrier core belongs to the finite PCR-3 comparison class;
2. the process has finitely many operational slots and a finite comb representation;
3. source, process, target, matching, boundary, continuation, standing-core, lawful-action, and physical-apex relations admit finite semialgebraic presentations over $\mathbb{R}_{\text{alg}}$;
4. every admitted physical amalgamation is specified by a finite typed gluing diagram and yields finite semialgebraic colimit and residual formulas;
5. arrows are lawful physical redescrptions preserving the complete enriched standing core.

No chart, chart index, chart certificate, successful overlap computation, or desired end-point appears in this membership condition. Write

$$\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF-fin}} := \mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF-fin}}, \quad \mathcal{L}_{D3}^{1.3,\text{RCF-fin}}(q) := \mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF-fin}} \times_{\mathcal{N}(T_{\text{osc}})}^h \mathcal{N}(T_{\text{osc}})[q].$$

The manuscript uses exact – fin as a short label for precisely this RCF – fin scope and for no larger certificate class.

The language is closed under finite conjunction, finite products, homotopy-fibre-product equations, projection, semialgebraic image and preimage, finite semialgebraic group actions, and stabilizer equations. Tarski-Seidenberg quantifier elimination therefore decides the resulting first-order formulas and supplies semialgebraic cells and sample points when nonempty. A nonalgebraic special branch does not enter this closure merely because one constructive witness is known. Until its certificates are proved closed under every pair, projection, stabilizer, and finite tuple construction, it receives NotReady or

$$\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}.$$

This is not elimination and does not alter its local target status.

Theorem 8.4 (STX-11: Actual chart-index typing and dispatch nonidentity). *Every actual chart has the form*

$$i = (a, \tau, H, \theta, W_H, \alpha_H, \kappa_H),$$

where a is a concrete PCR-5B/B15 frame, H is a finite source term, θ is an $\mathbb{R}_{\text{alg}}$ -semialgebraic cell, W_H is a finite comb, α_H is a physical apex, and κ_H is the proof object constructed from those data. The routing value $\tau \in \mathcal{D}_{12}$ is never chart identity, source identity, or overlap identity.

Proof. The fields are independently typed by the frozen B15 interface, PCR-3 source syntax, finite-operational process category, and semantic physical-apex relation. Dispatch records only the first target-facing H, U, L, or irreducibly joint work. Two objects may share τ while differing in any of $a, H, \theta, W_H, \alpha_H$, so the routing functor cannot be faithful and cannot serve as the atlas index. $\square$

9 Source-invariant framed normalization

9.1 Typed expression syntax

A source-invariant normal form requires every identifier-bearing mathematical expression to be typed before quotienting. The finite expression grammar has seven primitive leaf sorts:

1. source-node references $\text{SrcRef}(h)$;
2. parameter variables $\text{Par}(x:S)$;
3. operator or map symbols $\text{Op}(f:\sigma)$;
4. exact numerical coefficients in $\mathbb{R}_{\text{alg}}$;
5. target coordinates $\text{Tar}(t:S)$;
6. bound variables;
7. typed constants.

Application, indexing, finite tuples, Boolean and additive operations, and finite binders build compound expressions. The algebraic constructors are separately typed:

$\text{ScalarMul}, \quad \text{OrderedProduct}, \quad \text{Compose}, \quad \text{TensorProduct}, \quad \text{Link}.$

Only the scalar-factor list of ScalarMul is commutatively sorted, and every scalar factor must have an admitted commutative scalar sort. The remaining four constructors retain their operand order and declare an explicit result sort. A generic mul occurrence is ill-typed because it does not state whether the operation is scalar, ordered, compositional, tensorial, or a process link; it returns NotReady .

Under a lawful source isomorphism only SrcRef leaves move within semantic payloads. Parameter, operator, target, and constant identities are independently typed. Bound variables are serialized by de Bruijn index, so binder renaming is capture-avoiding. Every constitutive, residual, gluing, compatibility, witness, source-semantic, process, and target equation that can refer to source nodes must inhabit this grammar; an opaque string in such a field returns NotReady .

Reserved source-reference sorts. The expression sorts SourceNode , SourceRef , SrcRef , and their normalized source-reference aliases are reserved for the constructor source_node , which is the serialized constructor corresponding to $\text{SrcRef}(h)$. A parameter, constant, target coordinate, bound variable, call result, or algebraic result may not inhabit a reserved source-reference sort. Operator leaves do not admit a free sort field at all. A non- source_node expression that attempts to carry a reserved reference sort is ill-typed and returns NotReady . Thus a typed-expression label cannot provide a second route around SrcRef transport.

Total source-reference typing. Raw source-node identifiers are permitted only in the explicitly declared structural identifier fields: node declarations, relation endpoints, bearer/constituent fields, support declarations at parse input, and canonical relabel maps. Everywhere else a source-node reference must be represented as a typed $\text{SrcRef}(h)$ leaf. Raw semantic strings are typed only by their exact record-relative field/path schema. A string is accepted as ordinary nonreferential data only when that exact schema declares a non-source sort. An untyped or open semantic string fails closed regardless of whether its spelling presently equals any declared source identifier. In particular, inequality to the current source-node inventory has no typing force: adding, removing, or renaming an

inert source node may not change the type assigned to an existing semantic string. Every raw structural reference and every typed SrcRef is separately validated against the complete local plus B15-external source-node set before reduction; an unresolved reference returns NotReady.

Typed semantic map keys. Ordinary semantic dictionaries have exact closed key schemas. An undeclared key fails closed independently of its spelling and independently of the current source-node inventory. A source-indexed or otherwise open semantic map is represented instead by explicit typed key/value entries of the form `typed_map`; a source key is a `SrcRef(h)/source_node` expression, while a non-source key must carry an explicitly non-source semantic-key sort. Typed-map keys and values are transported under lawful source isomorphism and the entry list is then canonically sorted. An untyped dictionary key therefore cannot acquire source-reference authority.

Support desugaring. The auxiliary notation `support_refs` is eliminated before semantic reduction; it is not a side channel outside the source relation system. For each retained declaration that a source record owned by A is supported by B , normalization forms an explicit source incidence

$$\text{Support}(A, B).$$

For a multi-bearer record the same rule gives the finite typed support hyperedge from its structural owner set to the declared support node. This desugaring occurs before activity and spectator reduction. The resulting Support incidence participates in unknown-reference validation, activity adjacency, target-path closure, source-isomorphism canonicalization, and downstream explicit-incidence residual generation. The raw `support_refs` field is removed after desugaring; its source semantics survive only through the first-class incidence.

9.2 The B15-framed normal form

The scientific source input is the pair (a, H) , where a is the concrete B15 frame and H is the finite PCR-3/D3 constructor term. The normal form

$$N_{\text{src}}(a, H)$$

is the lexicographically least complete typed serialization among all type-lawful relabelings after inherited bearers have been materialized, source references validated, support declarations replaced by first-class incidences, typed expressions canonicalized, derivation hyperedges made explicit, and pointwise activity reduction performed. The reduction retains the complete constitutive target-path closure, including support and typed semantic source dependencies.

The preliminary node partition uses only identifier-free intrinsic data: constructor kind, source sort, fixed physical type or reference, and nonrelational attributes without source-node occurrences. Source-valued attributes, SrcRef leaves, constituent data, relational endpoints, equations, witnesses, and the desugared Support incidences remain in the complete relational payload. They are transported under each candidate relabel and enter only in the final full serialization. The mathematical normal form is defined by exact finite minimization; any bounded implementation that cannot complete that minimization has no positive or negative scientific force.

For

$$D = \text{Derived}(H_1, \dots, H_m; d, \mathcal{E}, b, f, p, w),$$

the normal form retains the derived node and

$$d : (H_1, \dots, H_m) \longrightarrow D,$$

together with typed constituent groups, map identity, typed constitutive equations $\mathcal{E}$, binding/fusion/projection data (b, f, p) , and typed source-semantic witness w . When constituent order is nonsemantic, both members and the complete group multiset are canonically sorted. When group positions are typed, their slot labels are semantic and the lawful automorphism group preserves those labels.

An inherited-bearer law obtains its bearers from a . A locally bearerless target-participating law must declare an explicit frame bearer scope. Such a record is retained even when it contains no local atom. Hence neither an inherited-bearer law nor a frame-scoped bearerless law can collapse to the empty source.

Definition 9.1 (Identifier-free preliminary partition). For every retained source node v , let

$$c_0(v) = (\text{ctor}(v), \text{sort}(v), \text{ptype}(v), \text{pref}(v), \text{intr}_0(v)),$$

where $\text{intr}_0(v)$ contains only intrinsic nonrelational attributes with no source-node occurrence. The following data are excluded from c_0 : input `support_refs` and their desugared `Support` incidences; source-node-valued attributes; every `SrcRef` expression leaf; constituent identifiers and groups; derivation, incidence, attachment, and process endpoints; target-participation and gluing relations; and every identifier-bearing equation or witness. These data remain in the complete relational payload.

Let $\Pi_0(a, H)$ be the finite partition by c_0 , and let $\text{Relab}_{\Pi_0}(a, H)$ be every type-lawful permutation inside its blocks. For $\sigma \in \text{Relab}_{\Pi_0}(a, H)$, transport the entire source payload, including every typed source reference, before serialization. Then

$$N_{\text{src}}(a, H) = \min_{\text{lex}} \{ \text{Ser}(\sigma_*(a, H)) : \sigma \in \text{Relab}_{\Pi_0}(a, H) \}.$$

An implementation bound that prevents exact enumeration returns `NotReady`, not a scientific source verdict.

Definition 9.2 (Typed algebraic operations). The expression grammar contains five distinct constructors:

$$\begin{aligned} &\text{ScalarMul}(\lambda_1, \dots, \lambda_r; X), & \text{OrderedProduct}(X_1, \dots, X_m), & \text{Compose}(X_1, \dots, X_m), \\ &\text{TensorProduct}(X_1, \dots, X_m), & \text{Link}(X_1, \dots, X_m). \end{aligned}$$

Only the declared commutative scalar factors λ_i are sorted. The distinguished value X , every ordered-product factor, every composition leg, every tensor slot, and every link-product slot retains its declared order. A bare `mul` constructor is ill-typed and fails closed.

Theorem 9.3 (STX-12R2: Transport-complete B15-framed normalization). *For every finite B15-framed source pair (a, H) , $N_{\text{src}}(a, H)$ terminates and is idempotent. If*

$$(a, H) \cong_{\text{src}} (a', H')$$

by a lawful typed source isomorphism, including transport through every source relation, semantic attribute, typed equation, and witness, then

$$N_{\text{src}}(a, H) = N_{\text{src}}(a', H').$$

The normal form preserves every active primitive or derived node, inherited bearer, derivation hyperedge, typed constituent group, incidence of every arity, attachment, source law, typed constitutive equation, binding/fusion/projection datum, physical reference, and typed source-semantic witness. Every semantic source reference is typed, raw

semantic strings are admitted only by an exact non-source field/path schema, semantic maps use closed key schemas or typed key/value entries, reserved source-reference sorts are inhabited only by SrcRef/source_node, every typed or structural source reference resolves before reduction, and every input support declaration is represented by a first-class Support incidence before activity reduction. Conversely, nonisomorphic derivation or support incidence is not collapsed, and an open string, undeclared semantic key, source-sorted non-SrcRef expression, opaque reference, or unresolved source reference cannot enter a solved normal form.

In addition, the B15 frame is classified exhaustively into source-semantic, target/standing, and pure-provenance fields before normalization. Only the source-semantic projection enters $N_{\text{src}}(a, H)$; an unknown top-level field fails closed. Source-node labels and source parameters are minimized jointly, with parameter colors determined by declaration metadata and with every typed occurrence in equations, incidences, and transport data retained in the complete serialization. If ψ_{Θ} is an identity-bearing alpha transport, the two framed pairs have the same N_{src} . For every other lawful invertible semialgebraic parameter transport, the verified source and target alpha canonicalizations conjugate it to a canonical normalized arrow

$$\psi_N : N_{\text{src}}(a, H) \longrightarrow N_{\text{src}}(a', H').$$

No lexical parameter-name ordering or size-dependent relabel fallback is admissible: resource exhaustion returns NotReady.

Proof. The constructor tree, frame bearer list, expression trees, semantic maps, and every constituent group are finite. Traversal emits finite records. Before reduction, semantic strings and map keys are checked against exact schema declarations, reserved source-reference sorts are checked against the expression constructor, structural references and typed SrcRef leaves are validated against the complete declared node set, and every input support declaration is replaced by a finite explicit Support incidence. String typing is completed before node-inventory resolution and never consults string equality with the node set; the latter is used only to resolve references already known to be structural or typed. Activity reduction removes only records that are source-null, explicitly inactive, or disconnected from every target-participating path; explicit derivation edges, support incidences, typed semantic source dependencies, and inherited bearers participate in that path closure. Consequently a support referent cannot be stripped as a spectator while the supporting relation remains semantically retained. The identifier-free partition $\Pi_0(a, H)$ has finitely many type-lawful relabels. Exact enumeration followed by complete-payload transport therefore has a lexicographically least serialization. Reapplication changes neither the schema typing, the reduced graph, the coarse intrinsic partition, the desugared support incidences, nor the canonical typed expressions, which proves idempotence.

Let $\phi : (a, H) \rightarrow (a', H')$ be lawful. Because c_0 contains no relabelable source identifier, ϕ induces a bijection between the blocks of $\Pi_0(a, H)$ and $\Pi_0(a', H')$, and hence a bijection between their candidate relabel sets. For each candidate, complete relational-payload transport relabels the desugared Support incidences, constituent data, all incidence/attachment endpoints, every typed-map source key, and every typed SrcRef leaf in semantic attributes, binding/fusion/projection data, gluing data, witnesses, lists, and dictionaries. No open semantic string, undeclared semantic key, source-sorted non-SrcRef expression, or unresolved reference reaches this stage: each has already failed closed by schema or reference validation. Since those decisions are schema-driven, the addition of a target-inert spectator cannot retag an existing semantic string. Bound names have already been replaced by de Bruijn indices, so the transport is capture-avoiding. Corresponding candidates therefore have identical complete serializations, the candidate sets coincide, and their least elements coincide. If instead the source type, fixed physical reference, explicit derivation incidence, or support incidence changes, no lawful candidate relabel identifies the resulting records.

For algebraic expressions, `ScalarMul` sorts only scalar factors of declared commutative scalar sort. The remaining four constructors preserve ordered argument lists. Therefore canonicalization cannot identify a noncommuting ordered product with its reversal, while it can identify a permutation of scalar factors in the same scalar action.

For the B15 frame, the schema first forms a disjoint three-way semantic partition. The joint canonical-label search is then performed on the complete pair consisting of the activity-reduced source hypergraph and the source-semantic frame projection. Parameter declarations are colored by their type and declaration metadata, while every incidence and equation occurrence remains in the serialized payload. Hence a mere parameter alpha-renaming induces a bijection of candidate joint labelings and has the same least serialization. A nontrivial lawful parameter transport is different: it is verified in the original charts and then conjugated by the chosen source and target canonical alpha maps. This produces ψ_N with exact forward and inverse semialgebraic expressions in canonical coordinates. Since the original inverse law is exact, alpha conjugacy preserves it. Thus the transport is retained as a genuine normalized groupoid arrow rather than being either collapsed by name sorting or lost from the source category. $\square$

Proposition 9.4 (Dual derivation and equation-alpha mutations). *Let H_A contain $d : A \rightarrow D$ and H_B contain $d : B \rightarrow D$, where A and B are source-type-distinct and physical-reference-distinct. Then*

$$N_{\text{src}}(a, H_A) \neq N_{\text{src}}(a, H_B).$$

By contrast, in the actual `idx-P-H1-B` chart the lawful relabel

$$\text{Le}_b \mapsto L_x, \quad H_b \mapsto H_x, \quad N_b \mapsto N_x, \quad N_{\text{mix}} \mapsto M_x$$

together with transport of

$$N_{\text{mix}} - P_{\text{mix}}(H_b, N_b) = 0$$

to $M_x - P_{\text{mix}}(H_x, N_x) = 0$ leaves the normal-form hash unchanged. Renaming a bound variable in a quantified constitutive equation also leaves the hash unchanged.

Proof. The first pair differs in a retained typed derivation endpoint. The second pair differs only by transport of `SrcRef` leaves; its operator symbol and all non-source leaves are fixed. The binder claim follows from de Bruijn serialization. $\square$

Proposition 9.5 (Relational alpha mutations). *Let H have two same-sort, same-physical-type nodes A, B , the symmetric retained relation*

$$A \rightsquigarrow B, \quad B \rightsquigarrow A,$$

represented in the input syntax by `support_refs` and therefore desugared before reduction to `Support(A, B)` and `Support(B, A)`, together with asymmetric target attachments $A \mapsto x$, $B \mapsto y$. Let ϕ be the lawful renaming $(A, B) \mapsto (Z, Y)$, transporting both relations and attachments. Then

$$N_{\text{src}}(a, H) = N_{\text{src}}(a, \phi H).$$

The same conclusion holds when the mutual source reference is carried by a `SrcRef` leaf inside a source-node-valued semantic node attribute.

Proof. Neither the input support declaration nor the source-node-valued semantic attribute enters c_0 . Support declarations have already become ordinary first-class `Support` incidences before activity reduction. Hence the original and renamed source pairs generate the same type-lawful candidate relabels. Complete-payload transport sends every support/incidence endpoint and every `SrcRef` leaf to the corresponding canonical identifier before lexicographic comparison. The two candidate-serialization sets are therefore equal. $\square$

Stability under source extension and relabeling. The typing rule is independent of accidental name coincidence. If a target-retained node A carries an open semantic field $\text{partner}=\text{"B"}$, normalization is undefined both when no source node B exists and when a target-inert spectator bearing that label is added. If the field is instead declared non-source, the same textual value remains ordinary data and the normal form is unchanged by the inert extension. Undeclared semantic keys, unresolved SrcRef leaves, unresolved support incidences, and attempts to inhabit a reserved source-reference sort through another constructor are likewise excluded. Typed source-key maps remain invariant under lawful relabeling. Conversely, an explicit $\text{Support}(A, B)$ incidence enters activity adjacency and target-path closure, so a genuinely supporting node B is retained. These are consequences of the source typing and normalization rules, not additional target selectors.

Proposition 9.6 (Ordered algebra hostile control). *For two noncommuting operator-valued expressions A and B ,*

$$\text{OrderedProduct}(A, B) \not\equiv \text{OrderedProduct}(B, A), \quad AB \not\equiv BA.$$

The analogous reversals remain distinct for Compose, TensorProduct, and Link. For scalar x, y , however,

$$\text{ScalarMul}(x, y; 1) = \text{ScalarMul}(y, x; 1).$$

A generic mul term is rejected.

Proof. The ordered constructors serialize their arguments in supplied order and perform no commutative sorting. ScalarMul validates each scalar factor against the declared commutative scalar sorts and sorts only that factor list. The constructors are disjoint in the typed grammar. $\square$

Proposition 9.7 (Unordered groups and law-only noncollapse). *Reordering the complete list of unordered constituent groups leaves $N_{\text{src}}(a, H)$ unchanged. Moreover, if H_{law} is either a frame-scoped target-participating law or a law on declared inherited B15 bearers, then*

$$N_{\text{src}}(a, H_{\text{law}}) \neq N_{\text{src}}(a, H_{\emptyset}).$$

In particular the permanent q_W source is not normalized to the empty source. A law-only source and a source-distinct local presentation can remain provenance-distinct while having a nonempty overlap over one standing core.

Proof. For an unordered group the canonical record is the sorted multiset of sorted member groups. An active frame-scoped law is itself a retained source record, and an inherited-bearer law additionally retains its typed ExternalBearer nodes. The empty source contains neither. Standing overlap is computed only after source normalization and does not require source equality. $\square$

Remark 9.8 (Supplementary conformance checks). The accompanying research archive contains finite checks of source-reference typing, alpha invariance, support incidence, ordered algebra, derivation fidelity, law-only noncollapse, and implementation fail-closed behaviour. These checks document conformance of the serialized representation to STX-12R2; they do not establish the theorem or alter its mathematical scope.

10 Standing-core equality and framed source provenance

For an accepted presentation x , the target-standing core signature $\Sigma_{\text{st}}^{1,3}(x)$ retains the old oscillation core, the represented process, applicable H/U/L statuses and values, QNorm,

QCov, matching, boundary, continuation, failure, lawful action, and target-facing common-apex data. It does not contain the complete source reduct merely because that reduct is retained for generation and audit.

For a standing core n , let

$$\widehat{\text{Prov}}_i(n)$$

be the framed provenance fibre carrying the B15 frame, normalized source term, construction history, source automorphisms, and frame-to-core map. A source coordinate enters $\Sigma_{\text{st}}^{1.3}$ only after an independent target activation theorem proves that the coordinate is standing-bearing.

Theorem 10.1 (STX-13: Standing-core/provenance separation). *For actual chart indices i, j , the scientific pair overlap is*

$$\mathcal{O}_{i,j} = \Gamma_i \times_{\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}}^h \Gamma_j.$$

Its objects are enriched standing-core arrows together with independent framed provenance data. Equality of source normal forms, construction histories, or source automorphism groups is not required unless an explicit activated source-tensor projection lies in the standing core.

Proof. SCIC identifies terminal carrier presentations exactly by the complete target-authorized standing continuation signature. Source provenance remains indispensable to candidate generation and later target extension, but stable reference forbids it from becoming a second terminal classifier without an activated target witness. The homotopy fibre product over $\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}$ enforces precisely the standing-core equality and lawful action. Each source presentation remains in its own framed fibre, so provenance is retained without being promoted. $\square$

11 The source-indexed Čech nerve

Definition 11.1 (Čech simplices). For $k \geq 0$, define

$$\mathcal{U}_{D3,k}^{1.3,\text{RCF}} = \coprod_{i_0, \dots, i_k \in \mathcal{I}_{D3}^{1.3,\text{RCF}}} \Gamma_{i_0} \times_{\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}}^h \dots \times_{\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}}^h \Gamma_{i_k}.$$

An object is a tuple of source presentations of one semantic standing core. The source terms remain separate; no source colimit, joint process, interference law, or physical amalgamation is inferred. Face maps forget a presentation factor. A degeneracy repeats the same actual chart object. Equal dispatch labels with distinct chart indices remain distinct factors.

Theorem 11.2 (STX-14: RCF-finite actual-pair decision theorem). *For each actual pair (i, j) in the RCF-finite chart domain, the complete semialgebraic standing-core arrow/equalizer formula and finite semialgebraic stabilizer formula have exactly one disposition*

$$\text{SolvedNonempty}(Z, \omega, \text{Iso}), \quad \text{SolvedEmpty}(c), \quad \text{Update}(u).$$

Within $\mathbf{QProc}_{D3}^{\text{RCF}-\text{fin}}$, NotReady does not occur. This decision problem neither compares nor amalgamates source reducts.

Proof. All standing-core coordinates, lawful actions, and parameter cells are finite semi-algebraic objects over $\mathbb{R}_{\text{alg}}$. The homotopy-fibre-product equations are therefore a first-order real-closed-field formula. Quantifier elimination decides emptiness and projection;

on a nonempty formula it returns a semialgebraic cell and sample standing-core arrow. The stabilizer is the semialgebraic subgroup preserving that cell and framed action and is extracted by the same exact equations. A first standing separator or inconsistency certificate proves emptiness. A changed target or primitive is an update. Source hashes and histories are carried in provenance fields and are absent from the equalizer unless independently activated. $\square$

Theorem 11.3 (STX-15: Direct higher Čech theorem). *For every actual finite tuple $(i_0, \dots, i_k)$, the direct common-standing-core formula is RCF-finite and exactly decidable. The resulting $\mathcal{U}_{D3, \bullet}^{1.3, \text{RCF}}$ is a complete simplicial groupoid. Distinct same-dispatch indices remain distinct; only literal repetition of one index supplies a degeneracy. No source colimit or X07/X08 residual is part of this theorem.*

Proof. The direct tuple is the iterated homotopy fibre product of the finite semialgebraic chart groupoids over $\mathcal{N}_{D3, \text{sem}}^{1.3, \text{RCF-fin}}$. Finite conjunction and projection preserve semialgebraicity, so quantifier elimination decides the direct tuple rather than inferring it from pairwise nonemptiness. Face and degeneracy maps are the canonical projection and diagonal maps of the iterated fibre product and satisfy the simplicial identities. Since source presentations are factors rather than identified source objects, equal dispatch labels do not collapse them. $\square$

12 Physical source amalgamation and its separation from Čech overlap

Definition 12.1 (RCF-finite physical-amalgamation object). An object of $\mathfrak{A}_{D3}^{1.3, \text{RCF}}$ is

$$\alpha_I = (I, D_I, \{\lambda_r : D_I \rightarrow H_{i_r}\}_{r \in I}, H_I, W_I, \alpha_I, \kappa_I),$$

where the λ_r are explicit typed source maps with gluing certificates, H_I is their activity-reduced source colimit, W_I is a declared finite process composition, α_I is one physical common apex, and κ_I is the complete RCF-finite residual and stabilizer certificate. Mere equality of standing cores supplies none of these maps.

Theorem 12.2 (STX-16: Typed source-amalgamation theorem). *The source colimit H_I is defined if and only if the physical-amalgamation input supplies an explicit typed diagram $D_I \rightarrow H_{i_r}$ for every member. Node names, dispatch equality, matching physical-reference strings, or nonempty Čech overlap do not create a gluing map. If a required leg is missing, the amalgamation has status NotReady; if a declared identification violates source typing, it has status SolvedEmpty; otherwise the finite activity-reduced colimit exists while retaining every node, derivation hyperedge, incidence, attachment, law, and member provenance not identified by the supplied diagram.*

Proof. A colimit is formed from morphisms, not from object labels. The explicit legs determine the only permitted node and relation identifications. Unidentified source nodes remain distinct. Since every member and gluing map is finite, the typed colimit is finite. Sort disagreement in one declared gluing class is a source-typing contradiction. Absence of a map leaves the scientific amalgamation undefined rather than licensing a disjoint-union substitution. Thus the construction neither confuses presentation overlap with physical co-instantiation nor silently changes the source architecture. $\square$

13 Complete joint residual structure

For a typed physical amalgamation, let $\mathfrak{R}_I$ denote its semantic PCR-4/D3 joint residual family. It is determined from the normalized source colimit and the finite-operational evaluator grammar before any auxiliary computation is considered.

Theorem 13.1 (STX-17R2: Transport-complete residual-generation completeness). *Every target-relevant semantic joint residual in $\mathfrak{R}_I$ belongs to exactly one primary generation class:*

1. *a normalized B15 source-domain condition defining Θ_a ;*
2. *a normalized B15 source equation defining $Z(E_a)$, an explicit retained source incidence, or an attachment;*
3. *a derived constitutive closure relation carried by an explicit DerivationMap;*
4. *an induced process-composition or interference relation;*
5. *a common-apex compatibility equation, including member restrictions, source-colimit typing, boundary, matching, continuation, gauge/anomaly/vacuum, and lawful-action conditions;*
6. *a typed target/source/process update.*

The generated family is sound and complete for the frozen finite PCR-3/D3 constructor and evaluator grammar. It is natural under every lawful source morphism $\psi = (\phi, \psi_\Theta) \in \mathbf{Src}_{D3}^{\text{coh}}$. Writing ψ_N for the canonical normalized arrow supplied by STX-12R2,

$$\mathfrak{R}_I(a', H') = \psi_* \mathfrak{R}_I(a, H).$$

This pushforward transports every normalized B15 source-domain condition and frame equation, every source-incidence residual, process/interference residual, and one-apex residual. For each member the compiled source locus is exactly

$$\Theta_a \cap Z(E_a).$$

Member parameters remain local unless an explicit parameter-gluing arrow identifies them. In particular, if coherent amplitudes A_i, A_j contribute to one observable channel, the generated formula contains

$$2 \operatorname{Re}(A_i A_j^*) = 2(\operatorname{Re} A_i \operatorname{Re} A_j + \operatorname{Im} A_i \operatorname{Im} A_j).$$

Proof. Proceed by structural induction on the complete normalized B15-framed source-amalgamation object. Every semialgebraic condition in `parameter_domain`, `source_parameter_domain`, `raw_source_domain`, or another schema-declared source-domain field generates class (1). Every frame equation is conjoined with those conditions before any decision. An Incide or Attach leaf contributes to class (2). A Derived leaf is represented by its explicit derivation hyperedge and generates class (3), including every stored constitutive and binding/fusion/projection relation. A process-law leaf generates its finite process restriction. Crossing a Compose or Hybrid constructor either composes finite combs, coherently adds amplitudes, or co-instantiates laws at one boundary; these are class (4). The polarization identity generates the cross terms from the component amplitudes rather than reading them from a pre-existing certificate. The one-apex evaluator grammar contributes class (5). PCR-4/D3 evaluator completeness states that every in-scope joint physical use factors through the saturated source/process/apex grammar. Therefore no additional same-scope residual channel remains. A proposed primitive outside that grammar changes the typed signature and is class (6), not an omitted residual.

Naturality is proved in the same induction. Every source incidence, derivation, and typed constitutive expression is transported functorially. Finite comb link products and polarization are polynomial operations, so they commute with the induced coordinate map. Target coordinates and common-apex equations are fixed, while member face maps are transported with their source chart. Thus the full residual family is the pushforward under ϕ .

For a general parameter transport, covariance is not inferred from alpha invariance. STX-12R2 supplies a candidate canonical arrow ψ_N , and STX-17R2 admits it only after the exact domain-relative obligations

$$\psi_N(\Theta_a) = \Theta_{a'}, \quad \psi_N^{-1}(\Theta_{a'}) = \Theta_a$$

and both inverse-composition laws have been proved. Every source polynomial or semi-algebraic residual F is then carried to $F \circ \psi_N^{-1}$; every target-side residual G pulls back to $G \circ \psi_N$. Exact inverse and domain laws give a bijection of the complete loci $\Theta_a \cap Z(E_a)$ and $\Theta_{a'} \cap Z(E_{a'})$. The physical source colimit namespaces member parameters and performs cross-member identification only when an explicit parameter-gluing certificate is present, so coincident printed names cannot create a false joint constraint. Since process composition, polarization, and one-apex equations are finite polynomial operations, substitution by ψ_N commutes with the complete residual grammar. This proves $\mathfrak{R}_I(a', H') = \psi_* \mathfrak{R}_I(a, H)$ for every lawful morphism of $\mathbf{Src}_{D3}^{\text{coh}}$, not merely for alpha-renamings. $\square$

Lemma 13.2 (Exact real-algebraic coefficient descent). *Let $\Phi(\mathbf{x}; \alpha_1, \dots, \alpha_m)$ be a finite first-order RCF formula whose coefficients $\alpha_j \in \mathbb{R}_{\text{alg}}$ are supplied by exact certificates*

$$(p_j(t), \ell_j, u_j), \quad p_j \in \mathbb{Q}[t], \quad \ell_j, u_j \in \mathbb{Q},$$

with exactly one real root of p_j in (ℓ_j, u_j) . Replace each α_j by a fresh real variable z_j and conjoin

$$p_j(z_j) = 0, \quad \ell_j < z_j < u_j.$$

Then Φ is equivalent, over every real closed field containing the certified algebraic coefficients, to the existential RCF formula over rational coefficients obtained by quantifying the z_j . The reduction is exact and presentation independent.

Proof. The minimal polynomial together with the rational isolating interval determines one and only one real root. Therefore an assignment of the original algebraic coefficient extends uniquely to an assignment of the fresh variable satisfying the certificate, and conversely every satisfying assignment of the rational formula assigns the certified root. Substitution gives a bijection of solution sets before any projection. Tarski–Seidenberg quantifier elimination then applies to the rational-coefficient formula. No numerical approximation is used. $\square$

Theorem 13.3 (STX-18R3: Semantic RCF reduction and standard quantifier-elimination closure). *For every finite physical-amalgamation presentation in $\mathfrak{A}_{D3}^{1.3, \text{RCF}}$, STX-17R2 constructs a finite conjunction $\Phi_{a,H}$ of typed polynomial equalities and inequalities over $\mathbb{R}_{\text{alg}}$, including the complete source domain Θ_a , the source equations E_a , the process and interference relations, the common-apex equations, the projected-cell comparison formulas, and the stabilizer-preservation formulas. Exact algebraic-coefficient descent produces a first-order formula $\widehat{\Phi}_{a,H}$ over $\mathbb{Q}$ with the same real solution locus. By quantifier elimination for real closed fields, satisfiability, emptiness, every required projection, projected-cell equivalence, and stabilizer conjugacy are exactly decidable and their solution sets are semialgebraic.*

If $\psi = (\phi, \psi_\Theta) \in \mathbf{Src}_{D3}^{\text{coh}}$ is a lawful source morphism, then STX-17R2 gives

$$\Phi_{a',H'} \equiv \psi_* \Phi_{a,H}$$

as formulas on the domain-relative semialgebraic isomorphism ψ_N . Hence the corresponding RCF solution loci are isomorphic, emptiness is preserved, projected cells transport, and stabilizers are conjugate. The induced arrow belongs to the semantic STX-20 Hom-groupoid by definition of lawfulness and by this covariance theorem. For the prompt calibration, the explicit typed legs and displayed physical point witness nonemptiness of the joint formula and yield a J1 chart over the common source/process apex.

Proof. STX-17R2 supplies a finite typed residual family and proves its naturality under the complete lawful source morphism groupoid. Replace every coefficient $\alpha \in \mathbb{R}_{\text{alg}}$ by a fresh real variable together with its rational minimal-polynomial equation and rational isolating inequalities selecting the intended real root. This preserves the represented real solution locus and gives a first-order formula over $\mathbb{Q}$. The Tarski quantifier-elimination theorem for real closed fields decides that formula and all existential projections; the Tarski-Seidenberg projection theorem keeps the resulting loci semialgebraic. Projected-cell equivalence and stabilizer conjugacy are instances of the same first-order decision theorem applied to symmetric-difference or counterexample formulas. Naturality of Φ under a lawful ψ_N transports the solution set and establishes the claimed groupoid arrow. The prompt witness is checked directly against the displayed finite formula, independently of any reference evaluator. $\square$

Corollary 13.4 (Mathematical authority of the RCF reduction). *The semantic conclusion of STX-18R3 depends on the mathematical reduction to first-order RCF and the standard quantifier-elimination theorem. It does not depend on a custom implementation, diagnostic token, conformance verdict, or reference computation. A supplementary computation may carry the diagnostic labels ReferenceSolvedNonempty, ReferenceSolvedEmpty, or AuditNotReady, but none of these labels is a premise of STX-18R3, STX-19, STX-20, STX-21, or STX-24.*

Proof. The theorem's premises are the typed semantic reduction, exact coefficient descent, lawfulness of the declared source morphisms, and the standard mathematical decision theorem for real closed fields. Computational records show only that a particular serialization, formula-formation path, or worked example was handled consistently. They have no independent admissibility-bearing or theorem-proving authority. $\square$

14 Semantic chartification and codescent

Theorem 14.1 (STX-19: Noncircular RCF-finite chartification). *For every*

$$X \in \mathcal{N}_{D3, \text{sem}}^{1.3, \text{RCF-fin}},$$

there exists an actual chart index

$$i = (a, \tau, H, \theta, W_H, \alpha_H, \kappa_H) \in \mathcal{J}_{D3}^{1.3, \text{RCF}}$$

representing X . Conversely, every actual RCF-finite chart represents an object of $\mathcal{N}_{D3, \text{sem}}^{1.3, \text{RCF-fin}}$. Chart possession is a conclusion of the theorem, not part of semantic membership.

Proof. The semantic object has a finite participating source core by definition of the physical comparison class, so PCR-3 source completeness supplies a source-coherent finite activity-reduced pair (a_0, H) ; STX-12R2 constructs the invariant framed source object $N_{\text{src}}(a_0, H)$ before chart membership is considered. Its retained oscillation restriction lies in the exact PCR-5B atlas, which supplies a concrete B15 frame a . The finite-operational process representation theorem supplies W_H , and semantic one-point physicality supplies

α_H . All source, process, target, action, and residual relations are semialgebraic over $\mathbb{R}_{\text{alg}}$; quantifier elimination and cell decomposition therefore produce a finite semialgebraic cell θ containing the represented point. The preceding source, process, restriction, standing-core, and cell proofs assemble κ_H . Dispatch routing assigns τ without deleting source syntax. Lawful alpha-isomorphisms yield the same framed normal form; every other lawful parameter transport yields the canonical normalized arrow ψ_N , which transports the chart cell, residual solution locus, and stabilizer naturally. Conversely, the chart data satisfy exactly the physical and operational membership conditions defining $\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}$. $\square$

For a connected base class q , let

$$\mathcal{P}_{D3}^{1.3,\text{RCF}}(q) := \text{codesc}\mathcal{U}_{D3,\bullet}^{1.3,\text{RCF}}(q),$$

and let

$$F_q^{\text{RCF}} : \mathcal{P}_{D3}^{1.3,\text{RCF}}(q) \longrightarrow \mathcal{L}_{D3}^{1.3,\text{RCF}-\text{fin}}(q)$$

forget chart presentation while retaining the semantic enriched object and lawful arrow.

Theorem 14.2 (STX-20 / GFC-09B^{1.3,exact-fin}: Exact connected-fibre reconstruction). *For every connected base class q , F_q^{RCF} is an equivalence:*

$$\mathcal{P}_{D3}^{1.3,\text{RCF}}(q) \simeq \mathcal{L}_{D3}^{1.3,\text{RCF}-\text{fin}}(q).$$

Proof. STX-19 supplies essential surjectivity from the independently defined semantic category through the alpha-invariant framed normal form. STX-17R2–STX-18R3 show that the chart formula and its semantic RCF solution object are natural under every lawful source-node isomorphism and every parameter transport in $\mathbf{Src}_{D3}^{\text{coh}}$. A semantic arrow is exactly a lawful enriched standing-core arrow together with the complete framed source action, including normalized nontrivial parameter-transport arrows ψ_N . These arrows are included in the semantically defined STX-20 Hom-groupoids rather than being discarded as presentation skin; the finite presentation layer supplies only explicit representations and conformance records. STX-13 identifies its chart-level representation with the homotopy fibre product over $\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}$, while STX-14 and STX-15 compute the complete pair and higher coherence groupoids. Conversely, every arrow of the chart-level Čech groupoid is required by definition to preserve the complete enriched standing core and the lawful framed-source action; a merely serialized record has no authority to create such an arrow. Hence the Hom-groupoids agree and the comparison is fully faithful and essentially surjective. $\square$

Theorem 14.3 (STX-21 / GFC-25^{1.3,exact-fin}: Exact global codescent). *The actual-index Čech nerve reconstructs the RCF-finite semantic category:*

$$\text{codesc}\mathcal{U}_{D3,\bullet}^{1.3,\text{RCF}}(q) \simeq \mathcal{L}_{D3}^{1.3,\text{RCF}-\text{fin}}(q) \quad \forall q \in Q_{\text{osc}}^{\text{SCIC}}, \quad \text{codesc}\mathcal{U}_{D3,\bullet}^{1.3,\text{RCF}} \simeq \mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF}-\text{fin}}.$$

Physical source amalgamations are represented as their resulting semantic J1 or other actual charts; they are not confused with presentation-overlap simplices.

Proof. The chartification theorem supplies an essentially surjective cover, and STX-14–STX-15 identify its complete Čech nerve over the semantic groupoid. The standard effective descent theorem for groupoids gives the codescent equivalence. STX-16, STX-17R2, and STX-18R3 separately ensure at the semantic level that any claimed physical co-instantiation is either a nonempty RCF-finite source amalgamation producing a semantic chart, an empty amalgamation, or an update. Thus mixed and higher source architectures enter the semantic cover as physical objects, not as illicit source colimits of presentation overlaps. $\square$

Mathematical dependence of the global closure

The global chain is

$$\begin{aligned} \text{STX-17R2} &\rightarrow \text{STX-18R3} \rightarrow \text{STX-19} \rightarrow \text{GFC-09B}^{1.3, \text{exact-fin}} \\ &\rightarrow \text{GFC-25}^{1.3, \text{exact-fin}} \rightarrow \text{STX-24}. \end{aligned}$$

Its mathematical authority lies in the semantic reduction to a finite first-order RCF formula, exact descent of real-algebraic coefficients, standard quantifier elimination, semialgebraic chartification, and groupoid codescent. Supplementary computations address representation fidelity, formula formation, provenance, and reproducibility only. The finite-operational QNorm argument, the STX-23 proper-continuum theorem, and the STX-25 role-occupancy conclusion are independent of this global-atlas chain.

15 A surviving prompt-HNL continuum

The published PCR-5D3 analysis constructs a compact prompt-HNL family indexed by $M \in [28, 30]$ GeV, with source-derived lifetime specialization and exact active block

$$N_M = D_M U_\nu, \quad N_M N_M^\dagger = \text{diag}(1 - \epsilon_p(M), 1, 1), \quad 0 < \epsilon_p(M) \leq 2.51 \times 10^{-6}.$$

The source lies in the finite-operational category, so STX-06 applies.

Theorem 15.1 (STX-22: Prompt-HNL survival under target 1.3). *The prompt-HNL family passes the 1.3 process-scope, QNorm, inherited QCov, finite-submatrix, low-mass LFU, H, L, matching, boundary, retained-base, and common-apex requirements. In particular,*

$$C_{ee}^{\text{op}, 1.3}(M) = 1 - \epsilon_p(M) > 0.9999949 > 0.987.$$

Its retained base classes $q_p(M)$ remain pairwise SCIC-distinct, so

$$\mathcal{Q}_{\text{prompt}} = \{q_p(M) : M \in [28, 30] \text{ GeV}\} \subseteq \text{im } \rho_{D3}^{1.3}, \quad |\mathcal{Q}_{\text{prompt}}| = \mathfrak{c}.$$

Proof. The source has finite quantum slots and Borel detector outcomes, so STX-02 represents it. On its factorizing finite-block branch, STX-06 identifies QNorm with the electron row of $N_M N_M^\dagger$, giving the stated strict margin. The inherited coverage calculation represents every U-active primitive in the QNorm, finite-submatrix, and LFU charts and keeps all other chart statuses at their previously proved Pass, N/A, or ROOS values. H and L results and the physical common apex are unchanged. Different M values change the physical threshold and oriented matching coordinate retained by the old packet, so the base classes are distinct. $\square$

16 The proper-continuum restriction image

Theorem 16.1 (STX-23: Proper-continuum restriction image). *For the full finite-operational semantic target,*

$$\text{im } \rho_{D3}^{1.3} = \{q \in \mathcal{Q}_{\text{osc}}^{\text{SCIC}} : \pi_0 \mathcal{L}_{D3}^{1.3}(q) \neq \emptyset\}.$$

It contains $\mathcal{Q}_{\text{prompt}}$ and excludes $\mathcal{Q}_{\text{sub}}$. Hence

$$\emptyset \subsetneq \text{im } \rho_{D3}^{1.3} \subsetneq \mathcal{Q}_{\text{osc}}^{\text{SCIC}}, \quad |\text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}, \quad |\mathcal{Q}_{\text{osc}}^{\text{SCIC}} \setminus \text{im } \rho_{D3}^{1.3}| \geq \mathfrak{c}.$$

The conclusion is independent of STX-20 and STX-21.

Proof. The displayed characterization is the defining image/fibre relation for the semantic restriction functor $\rho_{D3}^{1.3}$. STX-22 supplies a continuum subset of the semantic image. STX-10 proves directly that every full semantic fibre over $q_{\text{sub}}(\delta)$, $\delta \in [0.02, 0.03]$, is empty because the retained QNorm value is fixed below the mandatory threshold. Thus Q_{sub} is a continuum subset of the complement. Both inclusions are strict and the cardinal lower bounds follow. No source-atlas enumeration or codescent equivalence enters the argument. $\square$

17 RCF-finite closure

17.1 Closed RCF-finite sectors

The exact failure-route basis is

$$\begin{aligned} \mathcal{F}_{D3}^{1.3, \text{RCF}} = & F_{\text{src}} \cup F_{\text{proc}} \cup F_{\text{app}} \cup F_H \cup F_U \cup F_L \cup F_{\text{ret}} \\ & \cup F_{\text{match}} \cup F_{\text{bdy}} \cup F_{\text{phys}} \cup F_{\text{amalg}} \cup F_{\text{cross}} \\ & \cup F_{\text{quot}} \cup F_{\text{ov}} \cup F_{\text{upd}}. \end{aligned}$$

Its sectors close as follows:

Sector	Exact closure object
Semantic domain	Independent physical/operational definition of $\mathcal{N}_{D3, \text{sem}}^{1.3, \text{RCF-fn}}$; no chart certificate in membership
Source chartification	STX-19 construction from PCR-3, B15, finite combs, one-apex physicality, and RCF cell decomposition
Source normalization	STX-12R2 exhaustive B15 source-semantic projection, joint source-node/parameter canonicalization, total typed source-reference rule, first-class pre-reduction Support incidence, exact alpha collapse, and canonical normalized arrows ψ_N for every other lawful parameter transport
Process and applicability	STX-02–STX-06, frozen guards, QNorm 1.3, QCov, and finite-operational representation
Retained base	STX-07–STX-10 process-level factorization and complete-fibre elimination
Standing/provenance	STX-13 standing-core fibre products with independent framed source provenance
Čech overlap	STX-14–STX-15 direct RCF pair and higher standing-core fibre products; no source colimit
Physical amalgamation	STX-16 explicit typed gluing category; missing legs fail closed
Joint residuals	STX-17R2 complete typed grammar, B15 frame-equation inclusion, member-local parameter discipline, and full transport naturality $\mathfrak{R}_I(a', H') = \psi_* \mathfrak{R}_I(a, H)$
Semantic RCF closure	STX-18R3 exact $\mathbb{R}_{\text{alg}} \rightarrow \mathbb{Q}$ coefficient descent and reduction to first-order RCF; standard Tarski–Seidenberg quantifier elimination supplies satisfiability, projection, cell transport, and stabilizer conjugacy
Nerve/codescent	STX-20–STX-21 target-qualified GFC-09B/GFC-25 over the actual RCF-finite Čech nerve
Update/remainder	Explicit update, NotReady, or AtlasROOS $_{\text{fin-op} \setminus \text{RCF-fn}}^{1.3}$; no nonalgebraic witness is promoted into tuple-stable closure

Theorem 17.1 (STX-24: RCF-finite no-hidden-route and closure certification). *Within THUCL-D3-1.3 and $\mathbf{QProc}_{D3}^{\text{RCF-fn}}$, every additional same-scope D3 semantic object, distinction, presentation overlap, or physical source-amalgamation claim has exactly one disposition:*

1. *represented in the actual-index Čech codescent;*
2. *quotient-equivalent under the complete standing core with source difference retained in provenance;*
3. *a solved nonempty or empty RCF-finite physical amalgamation producing, or failing to produce, a semantic chart;*
4. *source-null, inactive, or target-inert under the faithful normalizer;*
5. *undefined or a typed update.*

No RCF-finite same-scope standing-bearing route remains outside the closed sectors. Therefore

$$\text{Status}_{\text{supp}}(D3, 1.3; \text{RCF} - \text{fin}) = \text{ClosureCertified}_{\text{RCF} - \text{fin}}^{1.3}.$$

For $\mathbf{QProc}_{D3}^{\text{fin-op}} \setminus \mathbf{QProc}_{D3}^{\text{RCF-fin}}$, the global-atlas status is $\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$, not ClosureCertified .

Proof. STX-12R2 closes the complete B15-framed source object: the frame has an exhaustive three-way semantic partition, source nodes and parameters are canonicalized jointly, alpha-renamings collapse to equal N_{src} , and every other lawful exact semialgebraic parameter transport survives as a canonical normalized arrow ψ_N . STX-02–STX-10 close finite-operational realization, QNorm, retained-base factorization, and the local target statuses independently of the global atlas argument. STX-13 blocks source provenance from acting as a hidden terminal classifier. STX-14–STX-15 compute the complete RCF-finite Čech nerve without introducing a physical source colimit. STX-16 and STX-17R2 place every normalized B15 frame equation, source relation, induced process/interference relation, and one-apex equation into one joint semialgebraic residual system and prove covariance under the full source morphism groupoid. STX-18R3 restores the declared $\mathbb{R}_{\text{alg}}$ coefficient scope by exact minimal-polynomial/isolation descent to rational RCF and invokes the standard RCF quantifier-elimination theorem to decide the complete conjunction, transport projected cells, and conjugate stabilizers. STX-19 proves noncircular chart coverage, and STX-20–STX-21 include every lawful normalized parameter-transport arrow in their semantically defined Hom-groupoids and identify the codescent; the serialized presentation is not the source of those arrows. SCIC and the complete 1.3 standing core exclude a second same-scope classifier. A non-RCF or tuple-unstable branch is outside the closure theorem and retains $\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$; it is not silently solved or eliminated. $\square$

Endpoint disposition

The scope-separated results are

$$\begin{aligned} \text{Type}_{\text{image}}(D3, 1.3; \text{fin} - \text{op}) &= \text{ProperContinuum}_{\text{fin-op}}^{1.3}, \\ \text{Status}_{\text{supp}}(D3, 1.3; \text{RCF} - \text{fin}) &= \text{ClosureCertified}_{\text{RCF-fin}}^{1.3}, \\ \text{Status}_{\text{atlas}}(D3, 1.3; \text{fin} - \text{op} \setminus \text{RCF} - \text{fin}) &= \text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}. \end{aligned}$$

The original THUCL-D3-1.1 Global Fibre Completion remains a separate unresolved programme; no Mini-ARC target substitution is made.

18 Role-occupancy consequence and downstream use

Theorem 18.1 (STX-25: Successor-target role-occupancy status). *If the successor target is evaluated as a one-terminal-complete-carrier role, its live eligible residue contains at*

least the continuum prompt-HNL base family and their accepted enriched lifts. Threshold and oriented-matching distinctions are standing-bearing and are not certificate-lawful role-incidence relabelings. Therefore the 1.3 successor certificate does not license a unique carrier realization.

Proof. STX-22 supplies continuum many pairwise base-SCIC-distinct accepted classes. A one-role target would require a unique compatible assignment orbit after the lawful quotient. Distinct retained threshold/matching values are target witnesses, not role-label permutations, so the assignments do not collapse to one orbit. $\square$

Status 18.2 (PCR-5E use). The successor target exports the full finite-operational proper-continuum image result and, separately, the RCF-finite source-indexed atlas, standing-coordinate ledger, source-composition data, and codescent object. The AtlasROOS^{1.3}_{fin-op\RCF-fin} remainder must not be treated as eliminated or as globally classified. None of these objects is silently substituted for the Mini-ARC’s currently frozen 1.1 D3 handoff.

19 Mathematical authority and computational verification

The proof and the accompanying computational materials occupy different logical roles. The semantic theorem chain consists of STX-17R2, exact algebraic-coefficient descent, STX-18R3, STX-19, the two target-qualified codescent theorems, and STX-24. Its decisive external input is the standard quantifier-elimination theorem for real closed fields, together with semialgebraic projection and ordinary groupoid descent. No claim of machine-checked formalization is made.

Proof-authority boundary

Computational validators are not semantic proof authorities. A failure of a reference implementation changes the status of that computation. It reopens a mathematical theorem only when it reveals an omitted semantic hypothesis, an invalid reduction, an erroneous inference, or a faithful same-scope counterexample.

The supplementary archive serves four narrower functions: it records target and programme identity; represents the typed source, transport, residual, projection, and stabilizer data entering the RCF formulas; checks the integrity of formula formation and selected conformance examples; and preserves provenance and reproducibility. Diagnostic labels such as ReferenceSolvedNonempty, ReferenceSolvedEmpty, and AuditNotReady have no independent theorem-proving force.

Accordingly, the public status ClosureCertified^{1.3}_{RCF-fin} denotes

- semantic reduction to the declared RCF-finite class
- + exact algebraic-coefficient descent
- + standard RCF quantifier elimination
- + chartification and groupoid codescent,

with the supplementary archive documenting scope, representation, provenance, and reproducibility. It does not assert that a custom program proves Tarski’s theorem.

20 Scope, limitations, and reopening conditions

Nonclaim 20.1 (No backward transport). The paper does not prove the THUCL-D3-1.1 Global Fibre Completion endpoint, does not supersede its approved skeleton, and does

not amend the Mini-ARC.

Nonclaim 20.2 (No unrestricted process or ultraviolet completeness). The parent category $\mathbf{QProc}_{D3}^{\text{fin-op}}$ is finite-operational rather than unrestricted infinite-dimensional or ultraviolet complete. The global source-atlas theorem is narrower still: it is proved only on $\mathbf{QProc}_{D3}^{\text{RCF-fin}}$. A point assigned $\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$ remains a live semantic parent-scope point with its local target status unchanged.

Nonclaim 20.3 (No provenance promotion). Recording a source term, physical-reference map, construction history, normal-form representative, or source automorphism group does not itself make that datum a terminal standing coordinate. Such data enter the core only through an independently activated target-facing projection.

Nonclaim 20.4 (No computational substitution). Finite conformance examples do not enumerate the scientific chart category and are not used to infer source completeness, pair completeness, higher completeness, codescent, or the proper-continuum endpoint.

The result reopens under a same-certificate counterexample to any load-bearing implication. In particular, it would be revised by:

1. a finite-operational law outside STX-02, or a positive normalized comb that fails to define the stated operational law;
2. a B15-framed source pair showing that STX-12R2 either deletes standing-relevant source structure, collapses nonisomorphic source terms, separates lawful alpha transport, admits an unresolved or untyped source reference, loses support-induced activity, or identifies noncommuting ordered operations;
3. an accepted same-base lift violating the retained-base equality $C_{ee}^{\text{op},1.3}(Y) = \overline{C}_{ee}^{\text{osc}}(q)$, or an accepted 1.3 lift over $q_{\text{sub}}(\delta)$ despite $1 - \delta < 0.987$;
4. an RCF-finite pair or higher tuple for which the stated Čech fibre product omits a lawful semantic arrow, introduces a nonsemantic arrow, collapses distinct actual indices, or forms a physical source colimit from presentation overlap alone;
5. a physical source amalgamation without the required typed gluing legs, or a valid amalgamation for which a retained source, process/interference, or common-apex residual is omitted;
6. a lawful parameter transport for which ψ_N fails to preserve the complete source locus, residual family, projected cells, stabilizers, or semantic Hom-groupoid;
7. an admitted algebraic coefficient for which exact minimal-polynomial and isolating data do not preserve the real solution locus, or an RCF reduction that omits Θ_a , an equation, projection condition, or stabilizer formula;
8. an object of the independently defined $\mathcal{N}_{D3,\text{sem}}^{1.3,\text{RCF-fin}}$ for which STX-19 supplies no chart, a chart that realizes no semantic object, or a mismatch between the semantic and chart-level morphisms used in codescent;
9. a finite-operational point outside $\mathbf{QProc}_{D3}^{\text{RCF-fin}}$ treated as globally solved or eliminated rather than assigned $\text{AtlasROOS}_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$;
10. a same-scope standing-bearing route preserving all closed sectors while changing the codescent object, or target formation that depends on the desired endpoint.

A Finite-operational representation

For finite-dimensional Choi spaces V_j , an operational law on instrument cones extends uniquely by multilinearity to a linear functional on $V_{n-1} \otimes \cdots \otimes V_0$. Complete positivity un-

der finite ancillary extension gives positivity of the representing comb. Substitution of deterministic channels at later slots yields the causal partial-trace recurrence. Standard-Borel outcome measures remain finite-dimensional operator-valued measures, and contraction with the bounded comb preserves countable additivity. These facts are the finite-operational representation used throughout Sections 3–6.

B Retained-base factorization diagram

The load-bearing square is

$$\begin{array}{ccc} \mathcal{N}_{D3}^{1.3} & \xrightarrow{C_{ee}^{op,1.3}} & [0, 1] \\ R_{D3}^{1.3} \downarrow & \nearrow \overline{C}_{ee}^{osc} & \\ \mathcal{N}(T_{osc}) & & \end{array}$$

and commutes before quotienting. Applying π_0 makes QNorm constant on every connected base class. The proof uses deterministic tester extension and link-product associativity, not a source-family-specific matrix identity.

C Actual-index and identity data

Field	Mathematical role	Identity status
a	complete PCR-5B/B15 frame	framed base provenance and chart domain
τ	H/U/L/J dispatch value	routing datum only; never chart identity
H	finite activity-reduced source term	source-semantic identity before standing descent
θ	exact source/process parameter cell	chart domain; may contain continuum many points
W_H	represented finite process	enters standing only through activated process/core coordinates
α_H	one physical-apex and common-cone record	target-facing standing when activated
κ_H	source, process, restriction, residual, and action proof data	proof object; its label does not create standing
$\Sigma_{st}^{1.3}$	complete terminal enriched standing core	sole terminal equality object at this target
$\widehat{\text{Prov}}_i(n)$	source identity, history, frame, and automorphisms over core n	retained provenance fibre, not a second classifier

D Normalization and decision structure

Operation	Mathematical action
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Typed expression transport	relabel source-node leaves in constitutive, residual, gluing, and witness expressions; preserve parameter, operator, target, and constant leaves; encode bound variables by de Bruijn index
Derived constructor	retain the derived node and an explicit DerivationMap with typed constituents, equations, binding/fusion/projection data, and source-semantic witness
Activity reduction	remove only source-null, inactive, or target-inert structure; derivation and support incidences remain in target-path closure
Framed source normal form	form $N_{\text{src}}(a, H)$, materialize inherited bearers, retain active law-only records, canonicalize unordered groups, and minimize the complete typed serialization
Čech pair or tuple	form the direct homotopy fibre product over $\mathcal{N}_{D3, \text{sem}}^{1.3, \text{RCF-fin}}$, retaining separate source provenance and never inferring a source colimit
Physical amalgamation	require explicit typed legs $D_I \rightarrow H_i$; form a source colimit only from those maps and retain unidentified source structure
Joint residual family	include source domains and equations, retained incidences, constitutive closure, process/interference relations, common-apex conditions, and typed updates; preserve naturality under lawful source transport
RCF decision step	reduce the finite typed formula to first-order RCF and invoke standard quantifier elimination and semialgebraic projection
Non-RCF remainder	retain NotReady or AtlasROOS $_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$; an isolated witness is not promoted into tuple-stable global closure

E Target and theorem summary

Object	Scope	Conclusion
THUCL-D3-1.1	original PCR-5D3/GFC	unchanged and separate
THUCL-D3-1.3	successor active closure	target evaluated in this paper
$\mathbf{QProc}_{D3}^{\text{fin-op}}$	finite-operational parent	retained-base factorization and proper-continuum endpoint
$\mathbf{QProc}_{D3}^{\text{RCF-fin}}$	RCF-finite semantic subcategory	chartification, connected-fibre reconstruction, global codescent, and ClosureCertified $_{\text{RCF-fin}}^{1.3}$
$\mathbf{QProc}_{D3}^{\text{fin-op} \setminus \text{RCF-fin}}$	parent remainder	AtlasROOS $_{\text{fin-op} \setminus \text{RCF-fin}}^{1.3}$, not elimination
STX-00-10	finite-operational chain	target separation, representation, normalization, retained-base factorization, and complete-fibre exclusion

STX-11-21	RCF-finite atlas chain	actual indices, source normalization, provenance separation, Čech overlap, physical amalgamation, residual naturality, RCF reduction, chartification, and codescent
STX-22-25	endpoint chain	accepted prompt continuum, proper-continuum image, RCF-finite closure, and non-singleton role occupancy

F Reproducibility note

The accompanying archive contains the target addendum, programme-separation memorandum, exact source and typed-semantic specifications, selected conformance examples, provenance records, and cryptographic manifests. These materials permit independent inspection of the represented target, formulas, and release identity. They are supplementary to the mathematical argument and make no claim of formally verified quantifier elimination or proof-assistant certification.

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